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Paleomagnetism of Small Basalt Exposures in the West Puget Sound Area, Washington, and Speculations on the Accretionary Origin of the Olympic Mountains

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Twenty-two sites from two small exposures of basalt of probable Eocene age immediately east of the Olympic Mountains have the following mean paleomagnetic direction: declination, 176.5°; inclination, −66.5°; circle of 95% confidence, 8.6°. This is the only paleomagnetic study of pre-Miocene rocks anywhere in western Washington or Oregon that matches the polar wandering curve for North America. However, a marked fanning of paleomagnetic declinations around the Olympic core suggests that concordance in this case is fortuitous and results from oroclinal bending during or immediately after Olympic orogenesis. Creation of the Olympic Mountains involved accretion of thickened basaltic crust followed by underthrusting of large quantities of oceanic sedimentary debris, as the Farallon plate pressed northeastward into a recess formed by the thick crystalline blocks of Vancouver Island and the North Cascades.

INTRODUCTION

Every early Tertiary rock unit in western Washington and Oregon so far studied paleomagnetically has a declination of remanent magnetism that points substantially east of the paleomeridian. This fact ordinarily is interpreted as evidence that large tracts of land, including the Oregon and Washington Coast Range (CR) and the Oregon and Washington Cascades, have rotated clockwise about a nearby vertical axis during Tertiary time. Evidence from the Oregon CR, in fact, shows quite clearly that rotation took place and, moreover, took place while CR rocks were in the process of formation [*Simpson and Cox*, 1977; *Beck and Plumley*, 1980; *Magill et al.*, 1981]. Thirteen separate studies involving at least 14 distinct rock units now support clockwise rotation (summaries in the work of *Beck* [1980] and *Magill et al.* [1981]). Thus it is of considerable interest that we have discovered several small exposures of early Tertiary volcanic rock east of the core of the Olympic Mountains that do not appear rotated. So common are clockwise rotations elsewhere, however, that we suspect that these concordant declinations conceal an unusually complicated history associated with the accretionary origin of the Olympic Mountains.

GEOLOGY

The two small areas of basalt we sampled are shown in Figures 1 and 4. Neither is more than a few tens of kilometers across, and neither seems to have been the subject of much detailed geological concern. The Port Townsend exposures (PT) generally seem to be lumped with the Crescent Formation, which is the large swath of Eocene basalt curving around the core of the Olympic Mountains (Figures 1, 4). The Bremerton exposures (BR) generally appear to have been ignored, at least in published reports. Both consist of aphanitic to microporphyritic flows ranging up to perhaps 10 m in thickness. Most are subaerial eruptions, judging from the presence of columnar jointing in many outcrops and the scarcity of pillow structures. Many of the flows are markedly amygdaloidal. Most are dipping, but the dips are gentle to moderate for the most part. The flows appear fresh and unaltered megascopically, except for some highly vesicular flow tops which have suffered considerable deuteric(?) alteration. The BR lava flows have been invaded by a small gabbroic plug [*Reeve*, 1979]. On the basis of their general similarity to other basaltic exposures in the Oregon and Washington CR, in terms of both lithology and geological setting, the BR and PT rocks probably are early or middle Eocene in age.

PALEOMAGNETISM

Exposures in the two volcanic areas we studied are very meager, and in situ, unweathered rock is difficult to find except in roadcuts, railroad cuts, and quarries. We managed to collect samples from 24 separate units, although results from two were rejected because of magnetic instability (see below). Sampling followed standard paleomagnetic practice, with in situ drilling and orientation by magnetic compass checked by sun compass or the back-azimuth technique. From five to ten separately oriented samples were collected at each site, depending on the extent of available outcrop; a space of at least 1 m (often 5 m or more) was maintained between samples. All samples were cleaned by alternating magnetic fields at peak value ranging from 10 to 50 mT. The optimum field for cleaning each site was selected by observing the response of two or more test specimens to progressive demagnetization to 100 mT or, in a few cases, even higher.

Most sites responded reasonably well to alternating field demagnetization, forming clusters with 95% confidence circles of 15° or less. This was the value we chose (arbitrarily) as a cutoff level; sites whose mean directions are less well defined than this probably add more noise than signal to the data set. Two sites were rejected on this basis, one from each outcrop area. In addition, in four instances samples were rejected in calculating site means on the grounds that they lay far outside a well-defined cluster of directions

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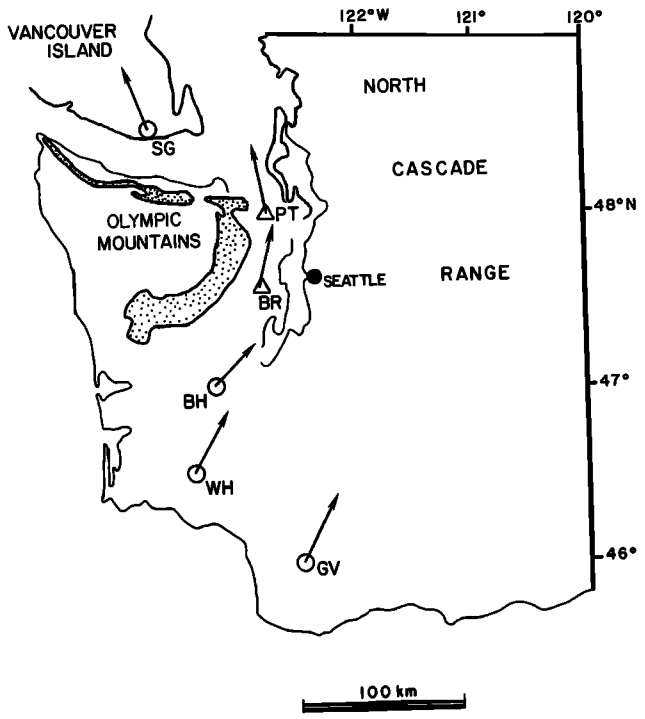


Fig. 1. Paleomagnetic results for early Tertiary volcanic rocks from the Coast Range of Washington and southern Vancouver Island. Arrows show rotation of declination ($D_o - D_x$, where D_o is observed declination and D_x is expected declination, calculated from *Irving* [1977]). An arrow pointing due north thus shows that the unit is unrotated. SG, East Sooke Gabbro; PT, Port Townsend basalts (this paper); BR, Bremerton basalts (this paper); BH, Black Hills Volcanics; WH, Willapa Hills Volcanics; GV, Goble Volcanics. All of these units are Eocene or probably Eocene in age except SG which is Oligocene. For references to published paleomagnetic results, see *Beck* [1980] or *Bates et al.* [1981]. Stippled pattern shows outcrops of the Crescent Basalt.

formed by the remaining samples from the same site. These samples probably represent cases of misorientation, or failure to recognize slumped or rotated blocks. Six samples (three from a single site) were eliminated by this criterion; all diverged from the mean direction (defined by the remaining samples) by at least twice the within-site angular standard deviation.

Table 1 gives paleomagnetic results for this study, and site mean directions are shown plotted in Figure 2. Table 2 gives expected directions for the two outcrop areas, and for the two areas combined, calculated from the reference poles of *Irving* [1977]. The 40 m.y. B.P. reference direction is plotted in Figure 2, for comparison with the mean direction for all 22 sites. Several characteristics of the data set deserve comment:

1. All sites have reverse magnetic polarity. We suspect this indicates that all the rock we sampled cooled during a single reverse polarity interval, although of course more complicated histories are possible. *Ness et al.* [1980] show several intervals of reverse polarity during the appropriate time range that lasted 1 m.y. or more, so magnetization during a single polarity interval does not necessarily imply magnetization over an unacceptably short period of time.

2. The BR and PT exposures have mean directions that are indistinguishable at the 95% probability level. Although the PT declination is considerably more easterly than the BR declination, steep inclinations for both groups bring their means close together. The two mean directions differ by 11.5°, which is of the same order of magnitude as the 95% confidence circles about each separate mean (Table 1). This closeness in direction, plus similar lithology, a similar structural setting, and a common magnetic polarity suggest that the two exposures are parts of the same volcanic sequence. Consequently, in the last line of Table 1 we lump all 22 sites and calculate a Bremerton-Port Townsend (BP) mean direction. The paleomagnetic pole corresponding to this direction lies at 87.5°N, 176.5°E, and has a 95% confidence oval with semiaxes of 11.5° and 14.0°.

3. Scatter in the BP data set is rather large. For instance, the between-site angular standard deviation for the entire data set is 21.3°, compared to 14.5° for the Goble Volcanics [*Beck and Burr*, 1979] and 16.8° for basalts of the Black Hills (B. R. Globerman and M. E. Beck, Jr., manuscript in preparation), both of which are collections of lava flows from nearby areas that are probably equivalent to the BP exposures in age. From the analysis of *Cox* [1970], values of 11°–15° are typical of the variability of the modern geomagnetic field at a latitude of 40°–50°. The additional scatter found in the BP data set no doubt mainly represents inaccurate tilt corrections. Tilts of up to 35° were measured in the field, but some of these measurements were scarcely more than approximations based on such insubstantial criteria as the general trend of columns and vague (but pervasive and consistent) flow banding. Reddened zones interpreted as flow contacts also were used in several instances, and these are notoriously suspect. Also, some of the 'tilts' may represent original dip, for which no correction should have been made. Nevertheless, scatter in the BP data set was reduced by making tilt corrections; the between-site angular standard deviation decreased from 30.2° to 21.3°, an amount that is statistically significant at the 95% confidence level [*Cox*, 1969]. The mean direction of all 22 BP sites without tilt correction is 148°, −68.5° (R = 19.02115). Owing to the steep inclination, this does not differ significantly from the corrected direction at the 95% confidence level.

4. The mean BP direction is virtually identical to the Eocene expected direction. This is shown in Figure 2, where the mean BP direction (inverted through the center of the projection) is compared with the 40 m.y. B.P. direction, calculated from the appropriate reference pole for the craton [*Irving*, 1977], using the dipole formula. Table 2 gives expected directions calculated from reference poles for 30, 40, and 50 m.y. B.P. [*Irving*, 1977]. Also shown are values of 'rotation' (of declination) and 'flattening' (of inclination) for BR, PT, and BP sets, together with their 95% confidence limits. Apparently, none of the three data sets is discordant in either declination or inclination, and this conclusion does not depend on precise dating of the rock. That is, the BP outcrops could be as young as mid-Oligocene or as old as early Eocene and the conclusion would still be the same—they have a direction of remanent magnetism that agrees with reference directions for the stable craton. Since every other pre-Miocene unit in the Coast Range and Cascades studied to date has had a discordant (clockwise rotated) declination, the BP rocks, although concordant, are clearly anomalous. Either the BP volcanics did not rotate clockwise during and/or after accretion, as did all similar exposures to the south, or they were rotated back to 'concordance' by a later tectonic event.

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TABLE 1. Directions of Remanent Magnetism in Volcanic Rocks Near Bremerton and Port Townsend, Washington

Site	N	R	H	D	I	α_{95}	DAZ	DPL	λ	ϕ
				Bremerton Area						
1	6	5.839	40	173.5	−62.5	12.2	80	17	47.55	122.71
2	6	5.952	20	180.5	−63.0	6.5	80	17	47.55	122.71
3	5	4.975	40	194.0	−53.0	6.1	80	17	47.55	122.71
4	4	3.954	10	153.5	−40.5	11.5	145	10	47.54	122.69
5	7	6.949	20	175.5	−57.5	5.6	145	10	47.54	122.69
6	3(1)	2.998	10	181.0	−56.0	3.9	145	10	47.54	122.69
7	6	5.945	20	158.5	−45.5	7.1	145	10	47.54	122.69
8	9	8.638	10	205.5	−65.5	11.2		flat	47.56	122.81
9	4(1)	3.980	40	190.0	−54.0	7.5	130	30	47.56	122.71
10	5(3)	4.781	40	183.5	−58.0	14.6	130	30	47.56	122.71
11	7	6.912	50	119.5	−82.0	7.4	89	28	47.55	122.71
12	7	6.859	50	277.0	−61.5	9.7	89	28	47.55	122.71
13	7	6.889	50	233.0	−65.0	8.3	89	28	47.55	122.71
Mean, BR	13	12.343		185.00	−65.5	10.0				
				Port Townsend Area						
1	7	6.930	20	189.5	−52.0	6.6	55	12	47.95	122.78
2	6	5.964	20	190.5	−38.0	5.7	55	12	47.95	122.78
3	6	5.964	10	127.5	−76.5	5.7	19	33	47.96	122.68
4	5	4.980	20	250.0	−83.0	5.4	19	33	47.96	122.68
5	6	5.968	10	139.5	−63.0	5.4	16	35	47.96	122.68
6	6	5.909	10	104.5	−54.5	9.1	350	25	47.96	122.68
7	8(1)	7.954	30	251.0	−68.5	4.5	295	10	47.90	122.73
8	9	8.979	10	126.0	−63.5	2.6	300	20	47.91	122.70
9	8	7.895	10	90.0	−74.0	7.5	0	30	47.94	122.69
Mean, PT	9	8.287		158.5	−71.5	16.1				
Mean, BP	22	20.494		176.5	−66.5	8.6				

N, number of samples; R, length of vector sum (of N unit vectors [*Irving*, 1964]); H, peak value of alternating field used in magnetic cleaning, in mT; D, I, declination and inclination of remanent magnetization, after magnetic cleaning and correction for tilt; α_{95}, radius of 95% confidence circle; DAZ, DPL, azimuth and plunge of dip vector used in making tectonic correction; λ, ϕ, north latitude and west longitude of sampling site.

INTERPRETATION

The geologic history of the Olympic Mountains is central to our interpretation of the BP paleomagnetic data. As described by *Tabor* [1972] and *Tabor and Cady* [1978*a*, *b*] the Olympics are an isostatically uplifted block of greatly thickened crust, formed by successive underthrusting of packets of oceanic sedimentary debris during subduction (Figure 3). As each 'packet' (sedimentary-tectonic unit) was emplaced, earlier arrivals were tilted continentward until the earliest are now vertical or even overturned. Ages young upward within each packet but, as a consequence of underthrusting, the packets themselves young downward. Rocks as old as Eocene and as young as Miocene are involved in the underthrusting. The abnormal thickness of low-density rock underlying the core of the Olympic Mountains gives rise to negative Bouguer gravity anomalies of up to −80 mgal, which contrast sharply with anomalies elsewhere in the Coast Range which are commonly +10 to +60 mgal [*Bonini et al.*, 1974; *Berg and Thiruvathukal*, 1967].

The arcuate pattern traced by basalts of the Crescent Formation around the core of the Olympic Mountains (Figure 1) has suggested to geologists for years that the Olympics may be an authentic, small-scale version of an 'orocline' [*Carey*, 1955], that is, that the curvature represents secondary (postorogenic or late-orogenic) bending about a vertical axis. Mapping of similar curvature in sedimentary-tectonic units and structures within the core of the mountains by *Tabor and Cady* [1978*a*] enhances this view. As much as 140° of bending is described by peripheral rocks (Crescent Formation and associated sedimentary rocks), whereas nearer 100° are found in sedimentary-tectonic units of the Olympic core. This apparent bending (curvature of map pattern) dies out to the west. 'Pencil structures' (thin slivers of rock defined by the intersection of several planar elements, such as cleavage or bedding) also are observed to fan around the Olympic core [*Tabor and Cady*, 1978*a*, Figure 24].

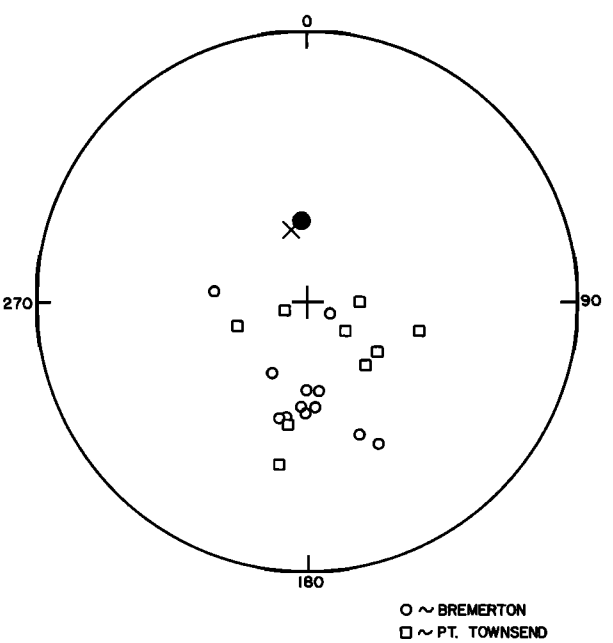


Fig. 2. Site mean directions of remanent magnetism for the Port Townsend (squares) and Bremerton (circles) basalt exposures. Solid circle is the mean for PT and BR data sets combined, inverted through the center of projection to give a normal polarity direction. Cross is the Eocene expected direction, calculated from *Irving* [1977].

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TABLE 2. Expected Direction of Remanent Magnetism, Clockwise Rotation of Declination, and Flattening of Inclination for Volcanic Rocks Near Bremerton and Port Townsend, Washington

Site	Expected Direction			Rotation			Flattening		
	30*	40*	50*	30*	40*	50*	30*	40*	50*
Bremerton (BR)	354/65.5	353/66.5	342.5/68.5	10.5 ± 23	12 ± 25	22.5 ± 24	3 ± 11	4 ± 12.5	5.5 ± 11.5
Port Townsend (PT)	354/66	353/67	343.5/68.5	−16 ± 61.5	−14.4 ± 62	−4 ± 61.5	−5.5 ± 17	−4.5 ± 17.5	−3 ± 17
Both (BP)	354/65.5	353/66.5	342.5/68.5	2.5 ± 23.5	3.5 ± 25.5	14 ± 24	−1 ± 10	0 ± 11.5	2 ± 10

Expected direction (D/I) calculated from reference poles of *Irving* [1977]. Rotation is observed declination (D_o) − expected declination (D_x); flattening is $I_x - I_o$. Error estimates are 95% confidence limits. For discussion of the statistical techniques, see *Beck* [1980].
*In m.y. B.P.

There exists an alternative to the orocline hypothesis, however. The Cresent volcanic 'horseshoe' could be a primary feature, and the apparent bending of sedimentary units within it could represent simple stacking of off-scraped oceanic debris against this curved buttress during subduction-related underthrusting [*Tabor and Cady*, 1978*a*]. This alternative lacks the episode of folding about a vertical axis that is essential to the orocline hypothesis.

In principle, it should be easy to distinguish between these possibilities, using paleomagnetic data from the Cresent Formation itself. The test involves 'unfolding' (rotation of the vector of remanent magnetism about the local strike through an angle equal to the local dip). After unfolding, oroclinally bent exposures will have magnetic declinations that fan around the Olympic horseshoe. Conversely, if the horseshoe bend is primary, after unfolding declinations will all point in the same direction. Seen in this light, the Olympic orocline problem is merely an extreme example of the problem of plunging folds discussed by *MacDonald* [1980], on a regional scale. Unfortunately, the test fails in this case, because the Crescent Formation stubbornly refuses to yield its direction of remanent magnetization! At least three attempts, in the form of full-scale paleomagnetic studies, have been made over the past 25 years, with no success (Allen Cox, personal communication, 1978; Sherman Grommé, personal communication, 1978; M. E. Beck, Jr., unpublished data). It seems that lava flows of the Crescent Formation have been prevasively altered, or lightly metamorphosed, leaving them megascopically fresh and promising, but magnetically worthless. Thus our BR and PT exposures, which appear from their magnetic stability to have escaped alteration but are close enough to the Olympic Mountains to have participated in Olympic orogenesis, assume a special significance.

As shown in Figure 1, magnetic declinations in Tertiary igneous rocks from Washington and southern Vancouver Island curve around the Olympic Mountains. This curvature suggests oroclinal bending. If declinations in the southern Washington CR give an 'undisturbed' direction, then bending ranges from about 16.5° eastward (clockwise) in the Black Hills to 47.5° westward (counterclockwise) in the East Sooke Gabbro. Total bending thus is about 65°, less than half the apparent bending of the Crescent Formation itself. The difference presumably might represent internal deformation within the Olympic block. Magnetic measurements thus are compatible with oroclinal bending—but they do not require it. Unfortunately, there appears to be no way to determine unequivocally whether curvature in map patterns in the Olympic Mountains and bending of magnetic declinations in units nearby are or are not causally related. Lacking a direct test, we instead examine the probable cause of oroclinal bending (subduction), and look for geological consequences.

D. C. Engebretson (manuscript in preparation) has made new calculations of plate interactions along the western margin of North America, based on marine magnetic anomalies. During the interval of interest here, the Farallon plate was underthrusting North America; from early Eocene through Miocene time the Farallon plate moved NE relative to North America, at a velocity of about 10 cm/yr. Both before and after that time velocities seem to have been more easterly, and larger.

As shown in Figure 4, this direction of relative plate motion could be responsible for bending of rock units in the Olympics and fanning of magnetic declinations further east.

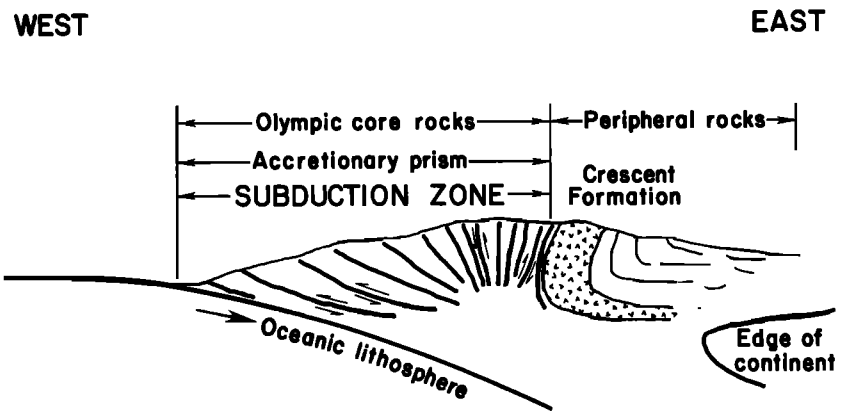


Fig. 3. Mechanics of subduction during Olympic orogenesis, taken from *Tabor and Cady* [1978*a*, *b*].

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Furthermore, if the Farallon plate were relatively 'warm' and buoyant, and therefore forcibly underthrusting North America [*Molnar and Atwater*, 1978], or if North America were advancing westward toward a trench that was anchored to the mantle [*Uyeda and Kanamori*, 1979], considerable compressive stress could have been exerted across the plate boundary. This might have bent the Crescent Formation and underthrust sedimentary units eastward, creating the Olympic horseshoe. Relative eastward movement of the Olympic block in turn could have been responsible for left slip on the Leech River fault of southern Vancouver Island [*Fairchild*, 1980]. Eastward encroachment of the Olympic block on stable North America could have been accommodated by underthrusting, crustal thickening, or downwarping—and this in turn may offer an explanation for substantial (at least 40–50 mgal [*Bonini et al.*, 1974]) negative residual Bouguer gravity anomalies in the Seattle-Everett area, directly opposite the eastward-bowing Olympics. Thus the chain of deductions and observations that supports the orocline hypothesis, although tenuous, hangs together. It remains to be tested by more detailed paleomagnetic studies of post-Crescent sedimentary rocks surrounding the Olympic horseshoe, and by work on Metchosin Formation lavas of southern Vancouver Island.

We should also comment on the difference between the tectonic history of the Olympic block and that of the Coast Range further south. Presumably after accretion of the CR block [*MacLeod and Snavely*, 1973; *Simpson and Cox*, 1977], subduction (northeastward underthrusting) was general along its western flank. Why then did the Farallon plate bend the CR block into a horseshoe pattern at its northern end, but not elsewhere? Similarly, why was underthrusting and crustal thickening concentrated so markedly in the Olympics? The answer probably lies in the existence of a curved buttress of thick pre-Tertiary crust bounding the Olympic Mountains on the north (Vancouver Island) and east (North Cascades). Sedimentary material delivered by subduction to the northern end of the Coast Range was in effect 'shoved in a corner.' Whereas south of the Olympics accreted material might move northward along the coastline (perhaps by strike-slip faulting), as well as downward into the subduction zone, at the northern end of the Coast Range block only underthrusting was possible. The supply of sediment from the continent also might have been abnormally great near the Olympics, as it is today, which would help to explain the increased crustal thickness. In our view, however, it was mainly the presence of the Vancouver Island–North Cascade 'corner' that made the difference. Basaltic crust delivered to the region by subduction crumpled northeastward against relatively unyielding crystalline crust to form the Olympic orocline. The fanning of paleomagnetic directions around the Olympic core was produced in the same way. Detachment of the thick basaltic accumulation (Crescent Formation) may have been made possible by the presence of a sedimentary horizon at its base (D. C. Engebretson, manuscript in preparation, 1982). Continued northeastward underthrusting then emplaced thick packets of oceanic offscrapings beneath and behind the Crescent horseshoe, causing crustal thickening, eastward and northeastward encroachment of the Olympic block on the craton (perhaps with further oroclinal bending), and eventually the Olympic Mountains by isostatic uplift.

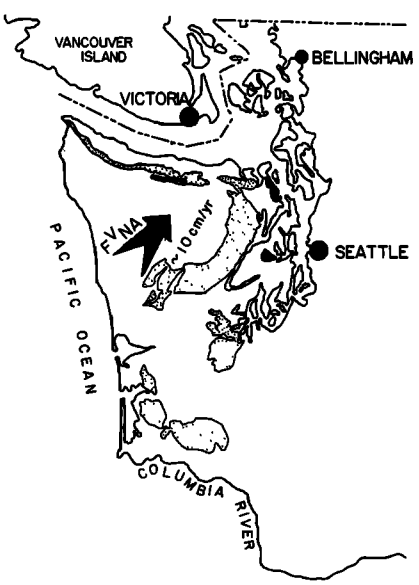


Fig. 4. Direction of underthrusting during Olympic orogenesis. PT and BR exposures are shown in black. $F^{V}NA$ is the velocity of the Farallon plate, relative to North America. After we had prepared this diagram, we discovered its remarkable similarity to Figure 27 of *Tabor* [1975], who also suggested that the curved trends in the Olympic Mountains have originated as oceanic material and was 'squeezed into a corner' between the Cascade and Vancouver blocks.

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